

JTCAM curation process with Solidipes curation tool

Guillaume Anciaux

JTCAM Data editor



Graphic from [PHD Comics](#)

FAIR Principles

What is **FAIR** (data) ?

- **Findable:** metadata, permanent links, searchable databases
- **Accessible:** Open and online services, archiving (long retention)
- **Interoperable:** standards and file formats and vocabularies/ontologies
- **Reusable:** Open (source) licenses, (software) context.

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Demands data curation

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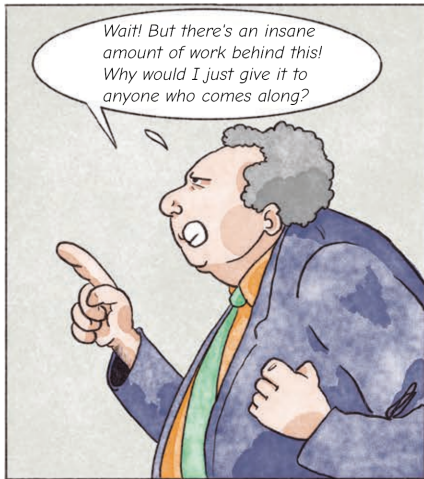
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Demands data curation

Definition: **Curation**

Organizing, cleaning, annotating dataset, ensuring relevance and reusability.

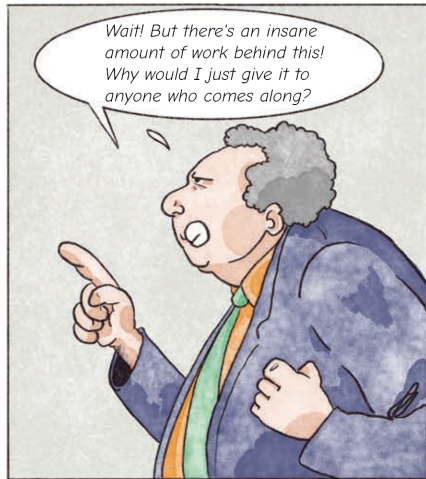
Why FAIR-ness ?



Translated from

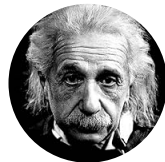
<https://zenodo.org/records/10108736>

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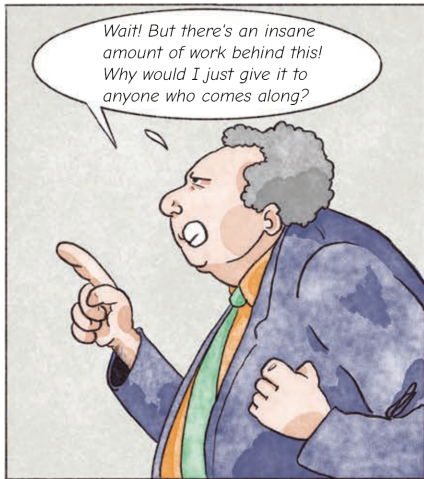
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*"The right to search for the truth
implies also a duty..."*

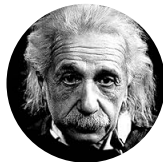
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*... one must not conceal any
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⇒ the created **network** when **sharing** data is the goal

JTCAM publishes curated data
accompanying papers

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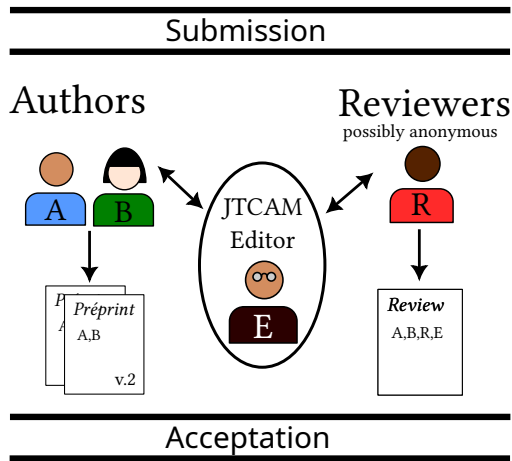
- Software Heritage (preserve code repositories)
- Zenodo (record < 50GB)
- Collects scripts Reproducing figures and/or results

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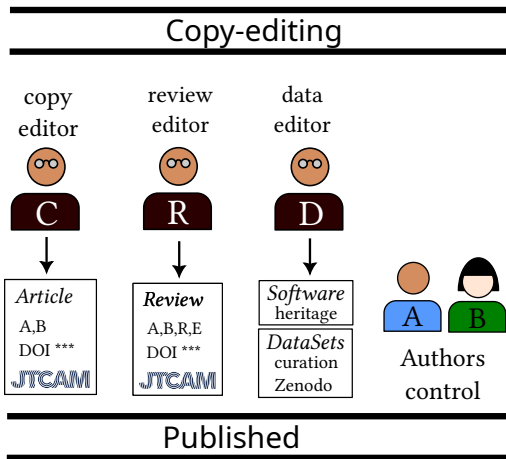
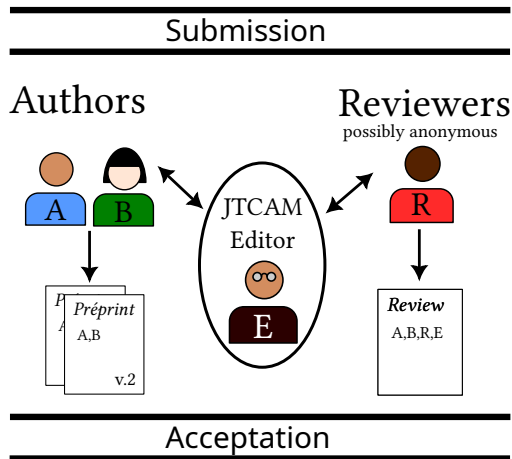
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using **dcsm.epfl.ch** online curation
dcsm $\Leftrightarrow$ *Overleaf* for datasets

JTCAM: publication process



JTCAM: publication process



Solidipes: analysis and curation tool



<https://gitlab.com/solidipes/solidipes>

Software funded by  **ETH BOARD** <https://ethrat.ch/en/>

Here, you can prepare your paper's data for publication in four steps:

✓ Acquisition

Upload any files relevant to your paper, and browse them like you would in a file browser.

Curation

Automatically verify the correct formatting of your files, review their contents, and discuss potential issues.

✓ Metadata

Easily edit any metadata relevant to your paper such as authors, keywords, description, and more.

Export

Once all previous steps are complete, review your work and export it to databases such as Zenodo.

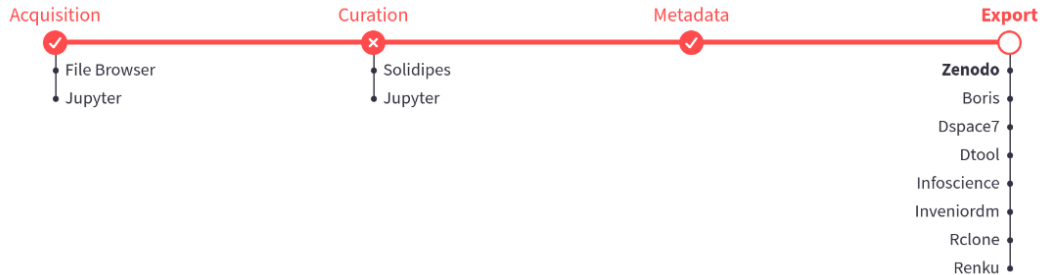
Solidipes: analysis and curation tool

Steps to collect curate and publish authors' data



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Solidipes Demo & Tutorial

Documentation: <https://solidipes.readthedocs.io>

Curated @JTCAM

1 Zenodo

E. Eid, R. Seghir, J. Réthoré.

Accompanying data for the paper "Crack branching at low tip speeds: spilling the T"

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data-science platform for reproducible science 



@renkulab.io

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Curation Tutorial

1 Installation

```
pip install solidipes
```

2 Downloading an existing dataset

<https://sandbox.zenodo.org/records/13642>

3 Reports

```
solidipes report curation/web
```

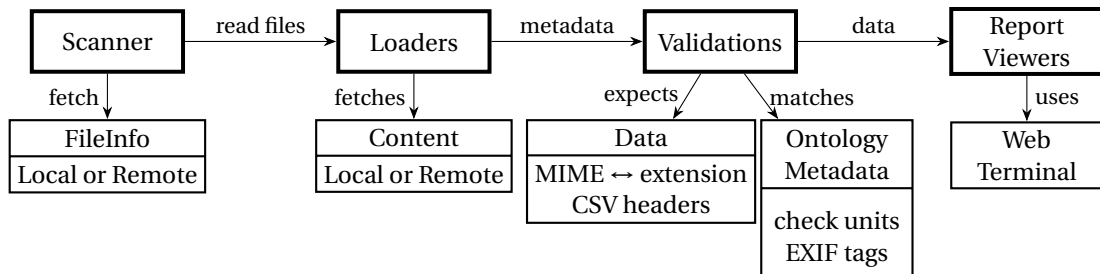
4 Tweak with plugins



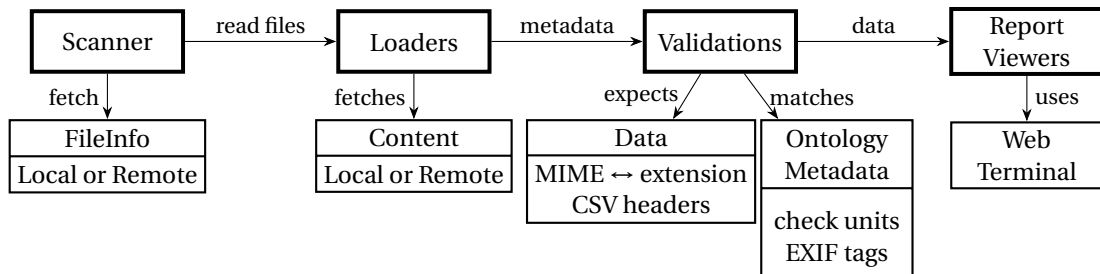
Supported Loaders/Viewers

 gnuplot	 image	 code_snippet	 yaml	 table	 json	 xml
 makefile	 hdf5	 zip_archive	 text	 matlab	 pdf	 notebook
 video	 python_pickle	 rdf	 FITS	 abaqus	 tikz	 pyvista_mesh
 meshio	 xdmf	 geo_msh	 geof_mesh	 binary	 dictionary	 pydantic

Plugins and Scientific Specializations



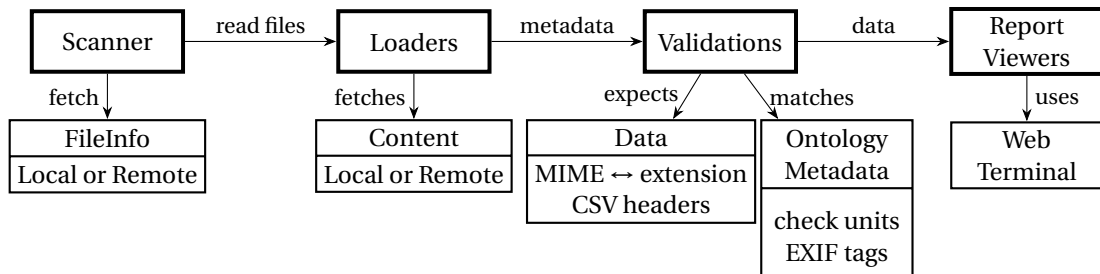
Plugins and Scientific Specializations



Template Plugin ⇒ Adapt

- Loaders
- Viewers
- Scanners
- Validators
- Ontologies (pydantic)
- Reports

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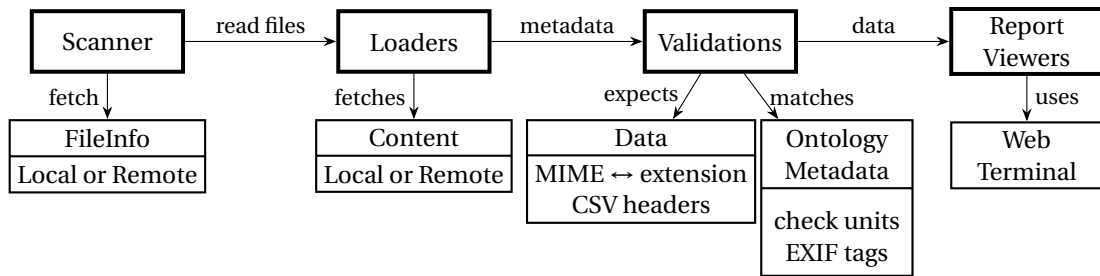
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Solid Mechanics Plugin

- Loaders: meshio, abaqus, paraview
- Viewers: 3D Web-rendering of meshes
- Ontology: check Displacement fields

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Astro Plugin

- Loaders: FITS
- Viewers: AladinLite, AstrovisJS
- Ontologies: TBD MModa
<https://odahub.io/ontology/>

Ontologies

Ontology (adapted from Wikipedia)

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In practice it is an **annotated graph** described with the *Resource Description Framework* (**RDF**) or as a set of *linked classes* (**Pydantic**)

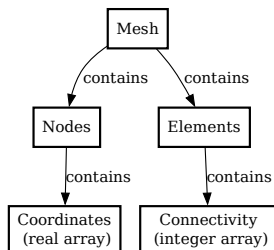
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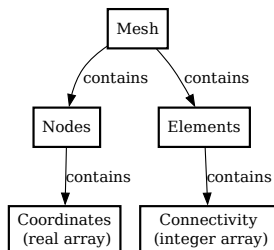
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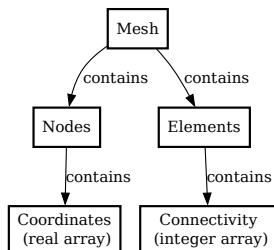
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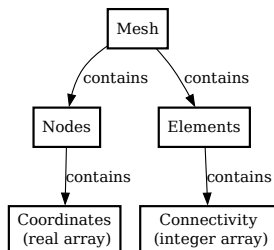
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- **Solidipes** stores metadata with **ROCrates** and validates with **Pydantic**

Reproducibility with workflows

Using workflow managers → binding to **DCSM+Solidipes**

Output: Crab [2026.07.08T21:39:58] ^ x

Query parameters Log Share ⓘ API code ⓘ **Open on Solidipes ⓘ** 

Showing 1 to 2 of 2 entries

Search:

Output Name	Image
picture_output	
spectrum_astropy_table	

Show 10 ▾ entries

Previous 1 Next

Multi Messenger Online Data Analysis(MMODA)
product Example

Conclusion

Open Science through Collaborative Data Curation

Motivation

- FAIR data (Findable, Accessible, Interoperable, Reusable) requires active **curation**.

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- **Curation:** *scan* → *load* → *extract metadata* → *validate* → *report* → *publish* (Zenodo, ...)
- **Remote storage** (Rclone, ...)
- **Domain specializations** (plugins, solid mechanics, astrophysics).
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Next steps

- **Workflows/containers** still work toward reproducibility (GUIX, Docker).
- **Reproducibility:** Author recommendations (**draft**)
- **Reproducibility:** Validations+Ontology ⇒ Quality badges

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